

Using OpenZeppelin for Developing Secured Smart Contracts

The blockchain has numerous applications but is susceptible to security lapses. OpenZeppelin helps integrate a security audit into blockchain-based algorithms called smart contracts.

Blockchains are today being used by government as well as corporate agencies because of the security, integrity and privacy they offer. These security features are provided by decentralised applications and smart contracts. Important applications of blockchain include cryptocurrencies, non-fungible tokens (NFTs), financial transactions, logistics management, e-governance, and many others.

Transaction logs associated with blockchain-based applications are secure and have no scope for hacking. Each and every record in the blockchain is linked to dynamic cryptography, enabling all transactions to be encrypted, thus eliminating the risk of sniffing or hacking.

The distributed ledger in a blockchain is a digital asset that has been copied, synced, and shared across several devices and locations to prevent manipulation by third parties. For instance, a bank can offer a better

level of security if it uses distributed ledger technology. The records of the transactions will be kept on one million devices if that bank has a million customers. So instead of just one server, the hacker will have to break into one million devices simultaneously. This is the main benefit of utilising decentralised blockchain technology.

If hackers gain access to a bank's server using a centralised application, they can copy all the customer information and data. That is the primary factor driving the decentralisation of web-based applications by government organisations.

Decentralised applications can be used to safeguard government servers that host land registry records, citizens' data (including Aadhaar in India), permanent account numbers (PAN), and so on.

Key application areas of the blockchain are:

- E-governance
- Banking and finance
- Stock market
- Insurance
- Internet of Things (IoT)
- Smart contracts
- Taxation
- Regulatory compliance and audit
- Voting/Polling/Elections
- Logistics management
- Cryptocurrencies and digital assets
- Non-fungible tokens (NFTs)
- Citizen identity management
- Money laundering protection
- Electronic health records
- Energy

